Re: TN266 v1p0 comments

Dealtry, Thomas [t.dealtry@lancaster.ac.uk]
Inviato: mercoledì 4 maggio 2016 16.30
A: Davide Sgalaberna

Hi Davide,

No need to send replies on the trivial stuff :)

Here's my comments on section 4 onwards. Line numbers correspond to the version you sent me.

~~Also something on Tab 12…~~
~~should be |dm13sq|~~
(I guess in most places you need to put the "|" around the dmsq's… a (careful!) global find replace would be useful. Perhaps also useful to do it for |dm32sq| (|dm13sq|) too)

Comment from Mark Hartz: We show the delta-chi^2 between the normal and inverted hierarchy cases, so I don't thing we are just measuring the absolute value of delta-m^2 .

~~Another thing that bugs me for no reason. Sometimes you have "run 1-6" and other times "Run 1-6". I prefer capitals, but best to just choose a style & stick to it. Same for Fig., Sec., Eq., etc.~~

Cheers,
Tom

~~527 - "All the systematics as well as oscillation parameters, which are nuisance, are marginalised" -> "All the systematics, as well as those oscillation parameters which are nuisance in a given measurement, are marginalised." - Just to clarify the statement. I think I'm understanding this correctly... e.g. when measuring dcp, you don't treat dcp are a nuisance parameter, but you do marginalise the other 5 (+ mass hierarchy? but according to previous sentence I guess not).~~

~~528 - Should link Tab 11 instead of Tab 12?~~

~~533 - N should be the number of Ereco * theta_l bins, shouldn't it?~~

~~537, 543 - shouldn't be starting a new paragraph on a "where x is …"~~

~~Eqn 7 - Capital A is undefined~~

~~560 - I'd take out the "(n = 10k)" here and stick a sentence at the end of the paragraph along the lines of "n = 10k is used for all results presented in this note; this value of n has been shown to be sufficient."~~
(you should link to proof of this. I guess there's a talk you gave you can reference?)

580 - sec. 3.1 -> Sec 3.1. I guess a global find/replace to make Tab., Fig., Sec., Eq., App., consistent should be done.

~~581 - You should clarify that the n=10k toys are generated for a given point on the theta-vector grid~~

~~592 - "minimum \chi^2 value, \chi^2_bf"~~

~~593 - discrete descent gradient algorithm should be referenced at the very least. Maybe even describe it.~~

~~605 - N^j is a function of a_i, not a?~~ actually it's only function of theta, since a is marginalized

~~610 - shouldn't be starting a new paragraph on a "where x is …"~~

~~624 - you should define what delta chisq is in the FC case (used in step 3, line 625)~~

~~625 - only step 2 is repeated (unless you split calculated fix, bf, and dchisq out)~~

~~625 - The two separate bits of maths next to each other confused me for a few seconds. I'd suggest "...each toy MC and f(dchisq), the distribution of dchisq, is obtained.~~

~~627 - "several critical values (e.g. for 68%, 90%, … CL)"~~

~~634 (and all bullets) - you mean Tab 11?~~

~~652, 654 - are the parameters not dm32sq (dm13sq)?~~

~~656 - does this mean that there are no "normal MH assumed" or "inverted MH assumed" results, but the results are shown as the best fit of normal or inverted at each point along the contour (which of course will jump between normal & inverted). Reading Sec 4.5, this isn't the case. So I'm not sure what you're trying to say here...~~ better explained in sec. 4.5

657 - surely for 1D dcp reactor, you use FC? no for that table we used const dchisq. will do with FC soon.

~~662 - "courser" -> "coarser"~~

~~689 - In the constant-dchisq case, if you draw a line @ dchisq = 1 (relative to the best fit mass hierarchy), the bits above the line are excluded @ 68% in the "global" sense, no matter if it's the normal or inverted hierarchy. So, I'd say that the constant values hold for the "best-fit mass hierarchy in the independent sense, and global" (in better words than that…) (What happens for FC, I don't know…)~~ it's probably not well explained. the thing is that const deltachisq cannot be used to exclude one mh from the other. So normal hierarchy contour is correct, while inverted one is wrong.

~~699 - add 68% = 1 (can use same sentence structure as 2D). Should also add the 2 sigma values for both 1D & 2D (you use 2 sigma in Fig 18). But of course, this is trivial from the Zsigma equation for 1D, so perhaps not. You decide!~~
~~Perhaps easier to just make a table with 1D, 2D vs 68% (~1 sigma), 90%, 95% (~2 sigma), 99%?~~

~~Fig 18 - this plot isn't a comparison with other analyses!~~

Fig 19 - watch out for the axis title overlapping the labels. e.g. y-axis of c,d, a-axis of all

Fig 19,20 - "68% CL" in the legend ("%%" is what you write in the code I think, depending on what method you're using to write the string)

~~703 - I'm actually confused about this… what I think is: (you have missed the \Delta's at the very least) "Fig 19 shows the sensitivity to sinsqtheta13 vs dcp and sinsqtheta23 vs dm32sq (dm31sq) with the reactor constraint. Fig 20 shows the sensitivity to sinsqtheta13 vs dcp with T2K data only. In both Figs. 19 and 20, normal and inverted hierarchy contours are set independently, i.e. each calculated with respect to its own minimum χ2 value and not the global minimum as a function of the mass hierarchy."~~

Fig 19,20 - I'm really confused… Make Fig 19 the plot with reactor constraint. Make Fig 20 the plots

with T2K data only. Make sure you write “T2K data only” in the legend of Fig 20. ~~Make sure the subfig captions are accurate (I’d put T2K only or reactor in the subfig captions too)~~

~~725,731,776 - extra paragraph break~~

~~729,790 - “continuos” -> “continuous” (try a global search/replace)~~

~~729 - “because periodic” -> “because it is periodic"~~

~~Fig 22 - could you repeat Fig 21(e) in Fig 22 (top right).~~ In the subfig caption, mention it’s repeated.

~~Fig 24 - again, maybe repeat Fig 21(e)~~

~~801, Fig 25 - “best-fit predicted spectra”. Also the legend - make it “best fit” instead of “expected"~~
This are expected spectra with Asimov data set A

Fig 25 - y-axis title

~~818 - helpful to have “-pi/2 = 1.571”~~

~~821 - You should reference Fig 26c~~

~~822 - how do you get the “if -pi/2 is assumed” from Fig 26c? This requires the 2D contour.~~

Fig 26 c) - add the yellow stripe to the legend

Fig 26 e) - put “ldm32sql or ldm13sql” as the x-axis title. Remember, all tech note plots are official plots, and can be used by collaborators without fully understanding them in context

~~Fig 27 - I guess that there is a “neutrino global minimum” and a “antineutrino global minimum”? If so, avoid the words “same global minimum”~~

830, Fig 28 - explicitly state that shape information is still used for the 1Rmu samples
The shape information is not used for the mu-like samples

Tab 25 - can you not add another interval column to the inverted hierarchy section. The 3 interval columns would then be labelled...
±1 sigma interval (independent & global)
±1 sigma interval (global)
±1 sigma interval (independent)
You’ll also have to change the text around 835

I don't understand. Since it's still not clear how to show the intervals maybe it's better to post-pone to a second time

Fig 29 - is the red-line here the same as the blue line in Fig 25 (for 1Rmu)? “Unoscillated prediction” may be better wording for the legend. Also, it looks slightly odd that the point in the legend is in the middle of 2 lines - move the Data to the top or bottom.

~~851 - goodness of test fit -> goodness of fit test~~

~~873 - shouldn’t be starting a new paragraph on a “where x is …”~~

~~878 - extra paragraph break~~

~~Tab 28, Fig 32 - you're linking eq 11, instead of the relevant section~~

~~Fig 32 - "and for the Run 1-6 data set." should be removed~~

~~887 - yeah, there is a slightly migration to high energy. But you should probably quantify it (i.e. it is small)~~

Fig 33,34 - same as some previous comments. Put % in the legend. Watch out for the axis overlaps. Put the yellow stripe in the legend. Remember to add (dm31sq) where relevant.

~~Fig 33, 893 - why are you using the word "global" here when you've defined it as the opposite as "independent"? It's just a bit confusing. Just remove the word "global" and you'll be fine.~~

Fig 34 - I don't think plotting the red star makes sense in this situation. When plotting globally, there is a single best-fit point that everything is calculated with respect to.

~~916 - if you want to have dchisq = 4.61 here (instead of 90%) then I'd suggest reminding people why, and even changing Fig 34 to use this notation. Or, just have 90% CL & 68% CL here.~~

~~950 - "significance to exclude dcp=0"~~

~~951 - "not excluded at 90% CL" -> "is excluded at less than 90% CL" (I'm not sure "excluded" and "not excluded" are trivial antonyms in statistical jargon. Perhaps get some advice from someone else on this)~~

~~952 - similar: The Run 1-6 data set excludes no CP violation in the leptonic sector at less than 90% CL.~~

957 - didn't you also say that there is a component from the fact that 1Rmu RHC has much lower stats (and so provides a worse constraint on theta23, dm32sq) than 1Rmu FHC? I don't recall seeing anything that separates this out from the FHC 1Re excess. In this case I mean the result of CP violation, though not explicitly written in the text --> added

~~Fig 36 - explicitly say that it is not a stacked plot~~

Fig 36 - the caption says there's more lines/colours than there actually are.

~~1100 - it's not strictly each flavour component… only the oscillated nue, oscillated nuebar, & everything else is grouped into "Bkg"~~

~~1106 - probably mean Figs. 37 and 38~~

~~Fig 37, 38, 39, 40, 41 - any chance of using subfig with captions for each? It's even more required for 40/41, since there isn't the writing in the figure~~

~~1113 - Mention that this a comparison of the breakdowns, rather than the total 2D distribution (which is done in Sec 6 somewhere)~~

~~Fig 40,41 - list of 7 plots in the caption, but only 6 shown. I guess you're not showing total?~~

~~1116 - Tab 38 listed twice. Guess it's 40?~~

~~Tab 37-40 - 102 systematics? Recall my comment from yesterday~~
~~What does "(oscillation excluded)" mean? They're in the table, so I guess it's the count of systematics~~

~~they're not in. Perhaps reword as 102 systematics + 4 oscillation parameters.~~
~~Mention in the caption that only the systematics that effect this sample are listed (e.g. RHC systematics aren't in the FHC table).~~

~~1118 - In Tabs. 41, 42, 43, and 44 the effect of each group of systematic uncertainties on the total number of events is shown for Asimov data set B. Tab. 45 shows the corresponding uncertainty on the ratio of neutrino and antineutrino mode events.~~

~~Tab 41-45 - recall my comment on the set A tables yesterday - try to make them clearer if possible~~
~~I guess I also had a comment in set A on "since theta23 and dmsq will be measured" - dcp and theta13 are being measured too!~~

Fig 43 - again, recall from yesterday. y-axis title should be "fractional uncertainty" or similar.
Think there was also comments on the caption of Fig 42/43 equivalent set A figure, that you should fix here too

~~1124 - I'm confused by this. The not-bias-corrected BANFF is used? Or no BANFF is used at all? If the latter, I don't understand how you got to the conclusion. You should also link to the "full BANFF corrected" results tables, and maybe compare some headline numbers to support the conclusion. For example, "The total errors are consistent when using the bias-corrected and bias-uncorrected BANFF results; the largest difference is x.xx% vs x.xx% in the 1R[mu,e] [FHC,RHC] sample"~~

~~Eqn 20 - mathcal{F} isn't defined anywhere (I think). Perhaps you mean "first the dchisq distribution, F(dchisq), is obtained"~~
~~You use a different notation (lower case normal mathmode f) in a previous section. I think they're the same thing - make them consistent.~~

~~1148 - code, you should use a different font style. Maybe try~~
~~https://en.wikibooks.org/wiki/LaTeX/Source_Code_Listings, but verbatim or textsc~~

> On 3 May 2016, at 23:12, Davide Sgalaberna <davide.sgalaberna@cern.ch> wrote:
>
> Hi Tom,
>
> I attached the updated note with Feldman-Cousins. This should be a version close to the final one.
> Just a comment: the coverage section is quite heavy in particular when the behavior of the deltachisq distributions of Feldman-Cousins is described.
> I think it's interesting to show the impact of the correlation between mass hierarchy and dcp but it's not easy to explain in a short and very clear way.
> Let me know if you think it's worth to put it in the backup.
>
> Your comments are not there in yet. I will address them tomorrow.
>
> Thanks!
>
> Cheers,
> Davide

Da: Dealtry, Thomas [t.dealtry@lancaster.ac.uk]

Inviato: martedì 3 maggio 2016 18.08
A: Davide Sgalaberna
Oggetto: TN266 v1p0 comments

Hi Davide,

Here's some comments on sections 1-3. Looking good - all just style/clarity points I think. I'll wait for the FC version before looking at the rest.

Cheers,
Tom

It's worth doing a global find-replace on super-K - should be Super-K

3 - ~~Steve is now Liverpool~~

~~45 - I guess you probably will/have, but remember to change the abstract when you add the FC results. I'd also quote some headline numbers here (e.g. exclusion region of dcp @ 90%, dm32sq and theta23 best fit & 1 sigma uncertainties, hierarchy preference, …)~~

~~89 - Missing full stop before "As in"~~

105 - Is there a sensitivity plot showing that 2D does nothing to help numu sample @ current stats? I'd reference this here. no there isn't. 2d templates for numu takes too much space

~~130 - "unique theta bin" -> "single theta bin"~~

~~Sec 2.1 - you say "lepton angle" everywhere, without distinction between "reconstructed lepton angle" and "true lepton angle". It's probably "reconstructed lepton angle" everywhere - just make sure & add it in~~
~~Also, what's beta in Eq 1?~~

~~Sec 2.2.4 - it'd be nice to have the variable cuts in the same form as 2.2.3 e.g.: (evclass == 1)~~

~~206 - Is it a fiTQun pi0 rejection cut?~~

~~303 - 14 10 degree bins from 0 to 140~~

~~306 - unique -> single~~

~~325 - don't reference table 9 (the SK error table) - it's not a BANFF output. Happens a few times~~

~~354 - Set A in Tab 12?~~

~~362 - 101-> 109 65+36+1+7(? not all 7 are systematic parameters at the same time of course, otherwise you wouldn't be measuring anything!)~~

~~377 - have a complicated -> have complicated~~

389 - there is no theta binning described
in http://www.t2k.org/asg/oagroup/inputs/201415/2015SKsplines

Tab 12 - "which corresponds to the opposite octant." makes me think it's a reflection into the opposite octant. But it's just an arbitrary point in the opposite octant. So perhaps "which corresponds to an arbitrary point in the opposite octant."

Tab 6-9 - can you add a column with things like f^banff_17;t;r to these tables? Because they're used in Tab 13 onwards without being explicitly defined. (This was a comment in my viva, otherwise I'd of never spotted it!)
Also, in the captions say something along the lines of "Only the systematic parameters which effect this sample are shown."
Careful with overflowing onto the page numbers too!

Fig 6 - dcp is measured too, isn't it? Maybe explicitly say that this is for the reactor constraint (the envelope will be slightly smaller in the T2K-only case, since theta13 won't be thrown)
continuos -> continuous
where is the blue (prefix)?
Fig 7 - as Fig 6 and...
y-axis title should be 'fractional error' (or similar)
Fig 8-12 - as Fig 6
Fig 13-17 - as Fig 7

Can you do a clearpage (or whatever) to put Tab 17-21 before Fig 8?

Tab 17-21 - this looks overly complicated… Can you put the XSec stuff first (it's the ugliest part) and maybe describe the things in more detail (what are the Correlated XSec parameters? Are MEC (corr) and MECbar (corr) included there?) Can you also choose whether you want the 'default' assumption to be pre-fit or post-fit, state that in the table, and only use the non-assumed one e.g. (pre-fit) on every line it's relevant
I'm also rather confused - sqrt(4.15^2 + 2.92^2 + 0.03^2) is not 11.9 they are not pre-banff
Style suggestion, use left alignment and some spacing to show what's a subset of what. For example (missing bits out…):
All
SKDet+FSI+SI
Flux+XSec
Flux
XSec Tot
XSec Corr Tot
MEC (corr)
NC 1gamma (uncorr)
XSec nue/numubar (is this corr or uncorr?)

<jointfit2016.pdf>